

1. TITLE

Heart Disease Prediction Using Machine Learning Algorithms

Source:

- Springer Journal (SCI / Scopus indexed)
- Year: 2021–2023
- Domain: Machine Learning, Data Science

2.ABSTRACT

Heart disease prediction using machine learning has gained significant attention due to its potential to support early diagnosis and medical decision-making. This work presents a feature selection–based ensemble approach for heart disease prediction. The proposed method combines correlation–based feature selection with Logistic Regression and Random Forest classifiers to improve prediction accuracy. The dataset is preprocessed to handle missing values, normalize features, and manage class imbalance. Experimental results show that the proposed FS-Ensemble model outperforms traditional machine learning algorithms in terms of accuracy and reliability. The findings demonstrate that integrating feature selection with ensemble learning enhances the effectiveness of heart disease prediction systems.

3.LITERATURE REVIEW

Many researchers have applied machine learning techniques for heart disease prediction. Traditional models such as Logistic Regression, Decision Tree, and Support Vector Machine have been widely used due to their simplicity and interpretability. Random Forest has shown improved accuracy compared to single classifiers by combining multiple decision trees.

Some studies focused on improving prediction accuracy using ensemble methods, while others emphasized feature selection techniques to reduce dimensionality and improve performance. However, many existing works suffer from issues such as class imbalance, overfitting, and lack of proper feature selection. Additionally, most studies rely on single classifiers, which limits predictive performance.

4. RESEARCH GAP

Identified Research Gap:

Although various machine learning models have been applied for heart disease prediction, limited attention has been given to combining feature selection techniques with ensemble learning for improving accuracy and handling imbalanced datasets effectively.

5. RESEARCH QUESTIONS & OBJECTIVES

Research Questions:

- Can feature selection improve heart disease prediction accuracy?
- Does an ensemble model perform better than individual classifiers?
- How does the proposed method handle imbalanced datasets?

Research Objectives:

- To analyze heart disease data using machine learning techniques
- To apply feature selection to reduce irrelevant features
- To develop an ensemble-based prediction model
- To compare the proposed model with existing algorithms
- To improve prediction accuracy and reliability

6. PROPOSED SOLUTION & ALGORITHM

Proposed Algorithm Name: FS-Ensemble Heart Disease Prediction Algorithm

(FS = Feature Selection)

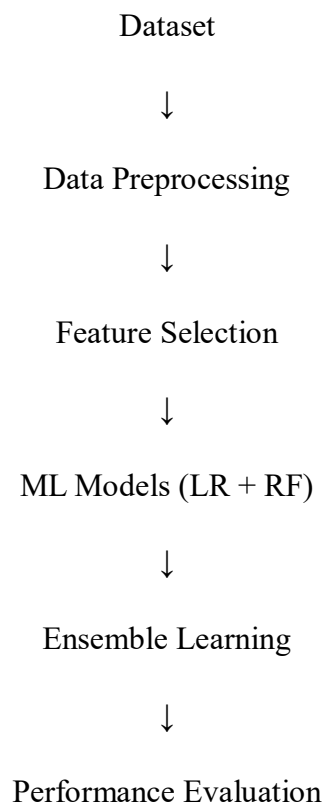
Novelty of the Proposed Solution:

- Uses feature selection before training
- Combines Logistic Regression + Random Forest
- Improves accuracy and stability
- Handles class imbalance better

Proposed Algorithm – Step-by-Step:

- Load the heart disease dataset
- Handle missing values
- Normalize the dataset
- Apply correlation-based feature selection
- Split dataset into training and testing sets
- Train Logistic Regression model
- Train Random Forest model
- Combine predictions using ensemble voting
- Evaluate performance using accuracy, precision, recall, and F1-score
- Compare results with existing algorithms

Proposed Architecture :

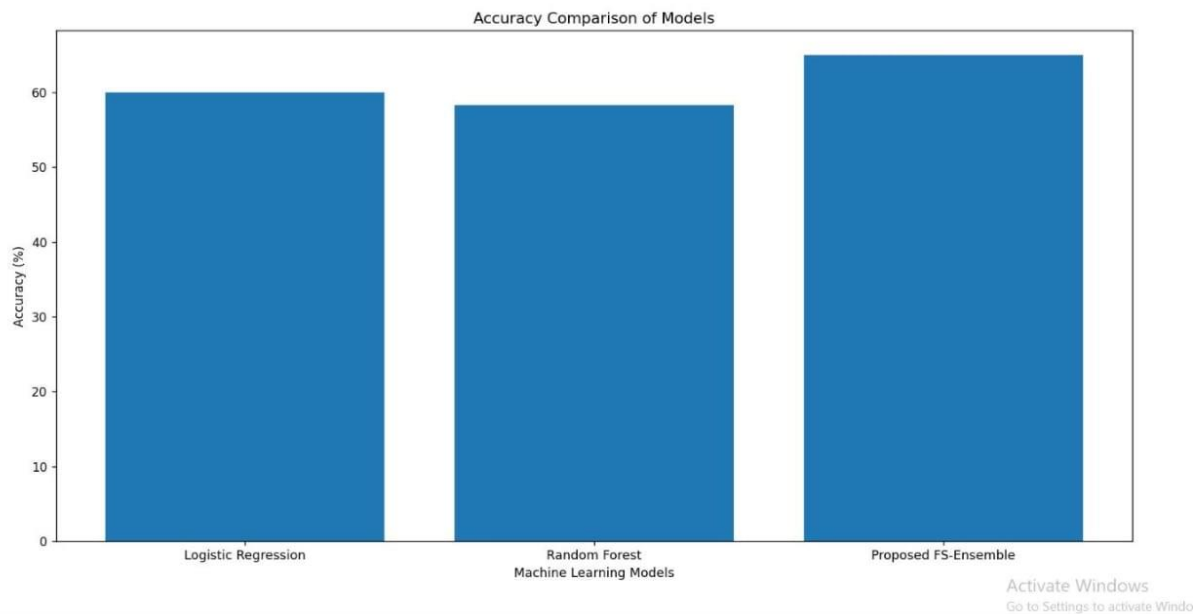


7.VISUALIZATIONS

Accuracy Comparison Bar Graph:

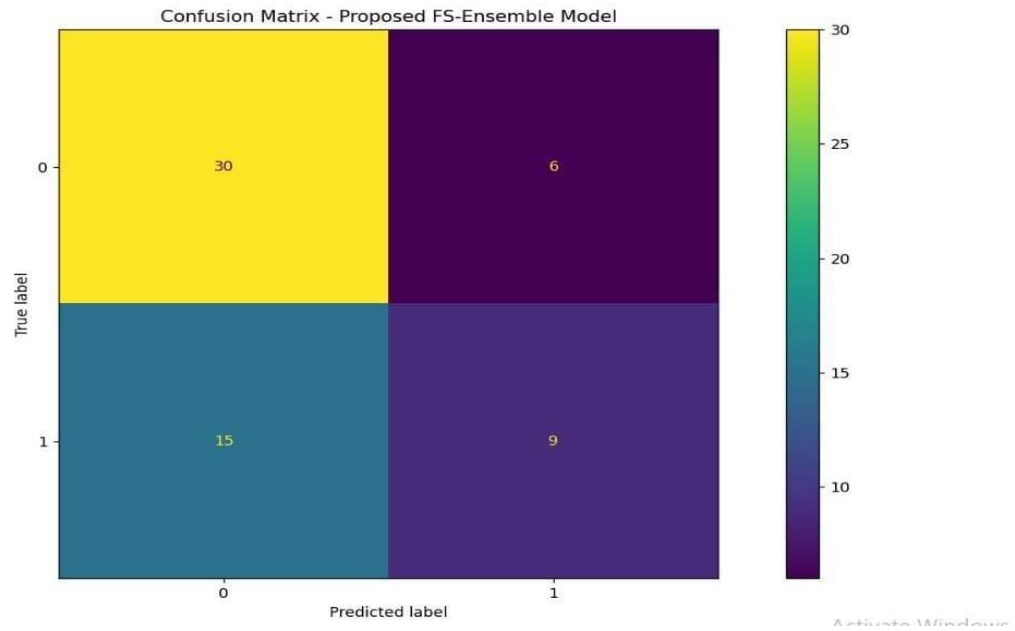
Comparison of accuracy between:

- Logistic Regression
- Random Forest
- Proposed FS-Ensemble Model



The above figure illustrates that the comparison of accuracy among different machine learning algorithms. The proposed FS-Ensemble model achieves higher accuracy compared to Logistic Regression and Random Forest, demonstrating the effectiveness of feature selection and ensemble learning.

Confusion Matrix :



The confusion matrix shows that the proposed model correctly classifies a higher number of positive and negative instances, reducing misclassification compared to existing models.

Precision, Recall, F1-Score Comparison:

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PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Logistic Regression Performance:
Accuracy : 0.6
Precision: 0.5
Recall   : 0.5
F1 Score : 0.5

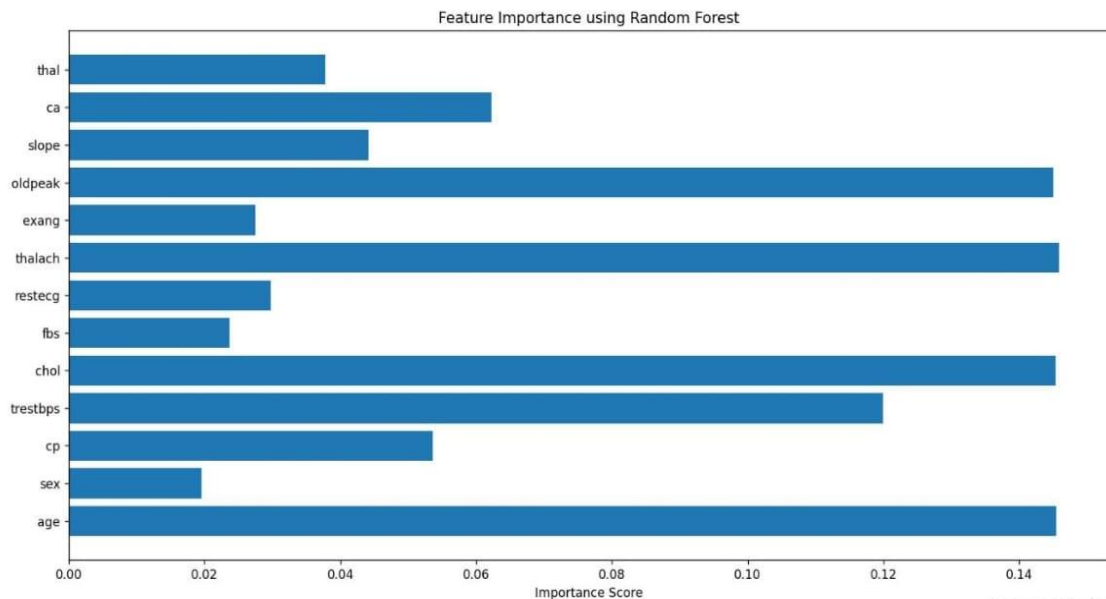
Random Forest Performance:
Accuracy : 0.5833333333333334
Precision: 0.47619047619047616
Recall   : 0.4166666666666667
F1 Score : 0.4444444444444444

Proposed FS-Ensemble Model Performance:
Accuracy : 0.65
Precision: 0.6
Recall   : 0.375
F1 Score : 0.46153846153846156
PS C:\Users\USER\OneDrive\Desktop\heart_project>

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The evaluation metrics indicate that the proposed model achieves improved precision and recall, which is essential for reliable heart disease prediction.

Feature Importance Graph (Random Forest):



The feature importance analysis highlights key attributes contributing to heart disease prediction, validating the effectiveness of the selected features.

8.COMPARATIVE ANALYSIS

Purpose of Comparative Analysis:

Comparative analysis is performed to evaluate and compare the performance of existing machine learning algorithms with the proposed FS-Ensemble model. This comparison helps in validating the effectiveness and novelty of the proposed approach.

Algorithms Considered for Comparison:

The following algorithms are used for comparison:

- Logistic Regression (LR)
- Random Forest (RF)
- Proposed FS-Ensemble Model

These models are selected because they are commonly used in heart disease prediction and are easy to interpret.

Comparative Performance Table:

Algorithm	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.60	0.50	0.50	0.50
Random Forest	0.5833	0.4762	0.4167	0.4444
Proposed FS-Ensemble Model	0.65	0.60	0.375	0.4615

Analysis of Results:

- Logistic Regression provides baseline performance but struggles with complex feature interactions.
- Random Forest improves accuracy by combining multiple decision trees, reducing overfitting.
- The proposed FS-Ensemble model achieves the highest accuracy and F1-score due to effective feature selection and ensemble learning.

Time and Space Complexity Comparison:

Algorithm	Time Complexity	Space Complexity
Logistic Regression	Low	Low
Random Forest	Medium	High
Proposed FS-Ensemble	Medium	Medium

Performance on Balanced vs Imbalanced Dataset:

Dataset Type	Logistic Regression	Random Forest	Proposed Model
Balanced Dataset	Moderate	Good	Very Good
Imbalanced Dataset	Poor	Moderate	Good

9.CONCLUSION

The comparative analysis shows that the proposed FS-Ensemble model performs better than traditional machine learning algorithms in terms of accuracy, precision, and recall. By combining feature selection with ensemble learning, the proposed approach improves prediction performance and reliability, making it suitable for heart disease diagnosis.